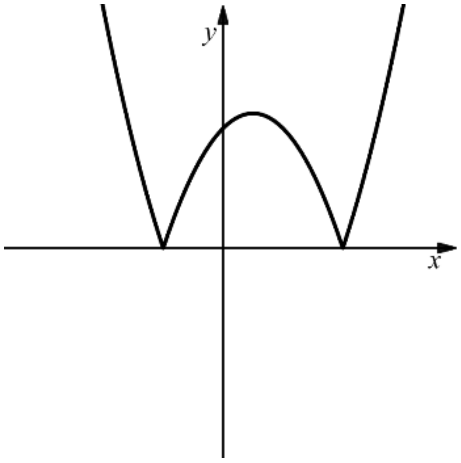
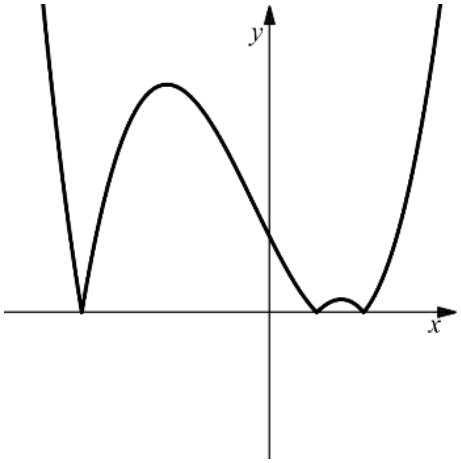
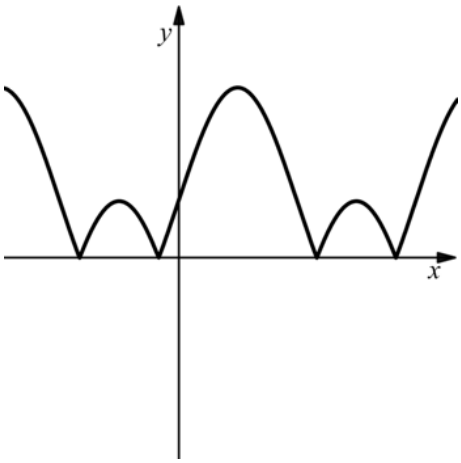
1) a)



1) b)

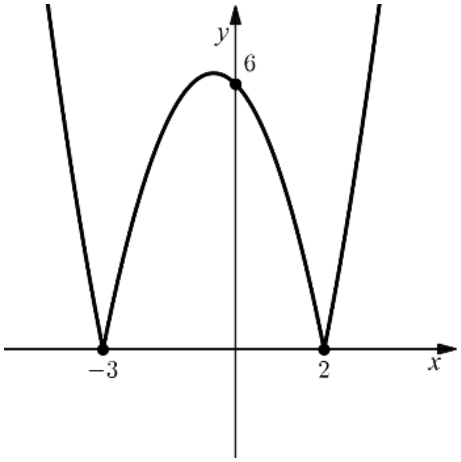


1) c)

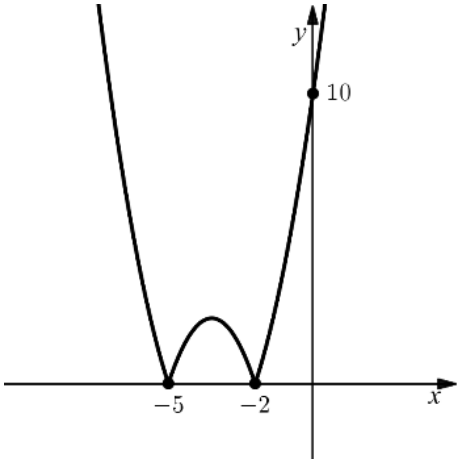


2) a)

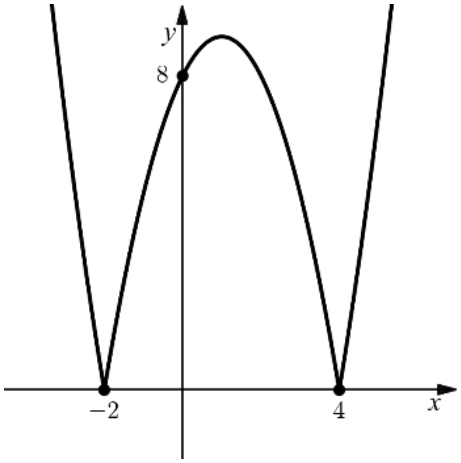
$x^2 + x - 6 = (x + 3)(x - 2)$



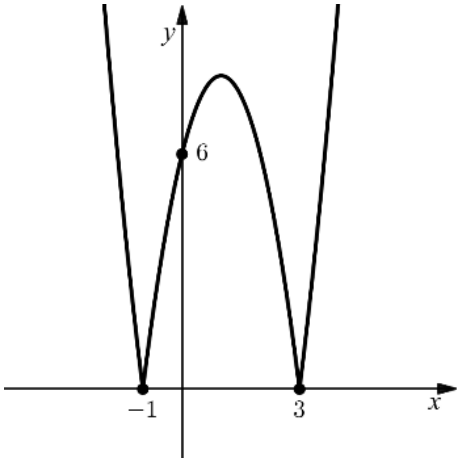
2) b)



2) c)



2) d)



3) a)

$k(x+1)(x-3)$ so $k=\pm 2$

$\pm 2(x^2-2x-3)$

$\therefore a=2$, $b=-4$ and $c=-6$ or $a=-2$, $b=4$ and $c=6$

3) b)

$kx(x-5)$

$k\frac{5}{2}\left(\frac{5}{2}-5\right)=\frac{25}{4}$ so $k=-1$

$\pm(x^2-5x)$

$\therefore a=1$, $b=-5$ and $c=0$ or $a=-1$, $b=5$ and $c=0$

3) c)

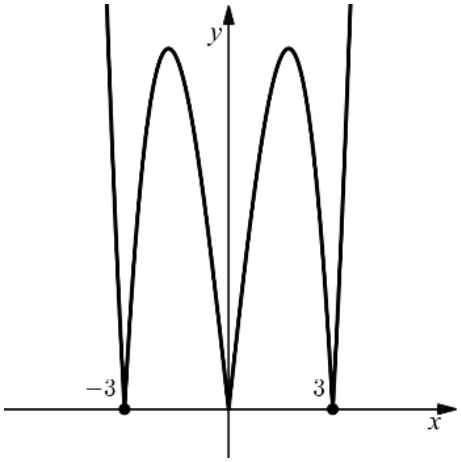
$k(x-2)^2$ so $k=1$

$\pm(x-2)^2=\pm(x^2-4x+4)$

$\therefore a=1$, $b=-4$ and $c=4$ or $a=-1$, $b=4$ and $c=-4$

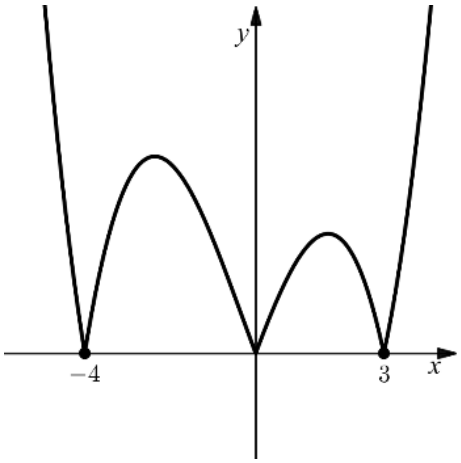
4) a)

$x^3 - 9x = x(x-3)(x+3)$



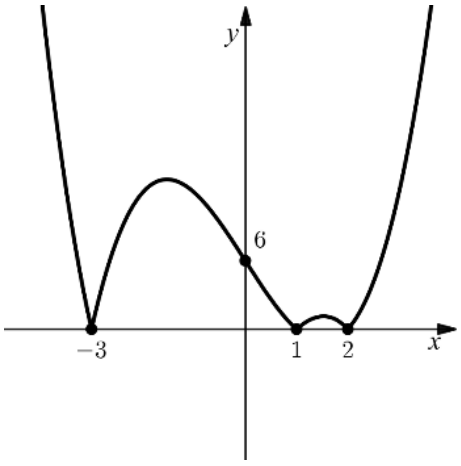
4) b)

$x^3 + x^2 - 12x = x(x+4)(x-3)$



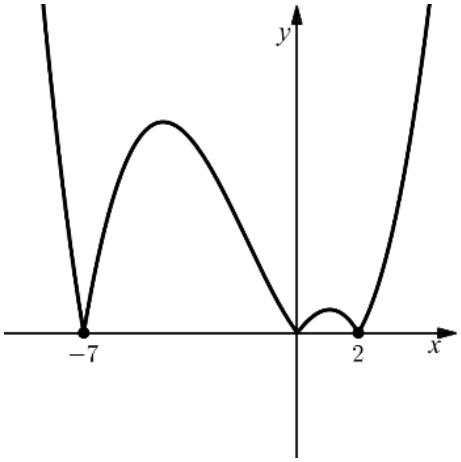
4) c)

$x^3 - 7x + 6 = (x-1)(x-2)(x+3)$



4) d)

$14x - 5x^2 - x^3 = -x(x+7)(x-2)$



5) a)

$k(x+3)(2x+1)(x-5)$ so $k = -1$

$\pm(x+3)(2x+1)(x-5) = \pm(2x^3 - 3x^2 - 32x - 15)$

$\therefore a = 2,\ b = -3, c = -32$ and $d = -15$ or $a = -2,\ b = 3, c = 32$ and $d = 15$

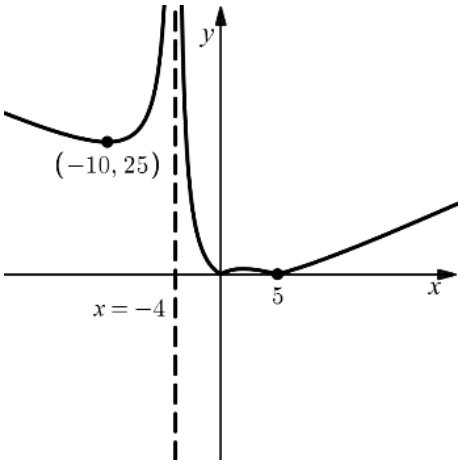
5) b)

$k(x+4)(3x+1)(x-2)$ so $k = -\frac{1}{2}$

$\pm\frac{1}{2}(x+4)(3x+1)(x-2) = \pm\left(\frac{3}{2}x^3 + \frac{7}{2}x^2 - 11x - 4\right)$

$\therefore a = \frac{3}{2},\ b = \frac{7}{2}, c = -11$ and $d = -4$ or $a = -\frac{3}{2},\ b = -\frac{7}{2}, c = 11$ and $d = 4$

6)

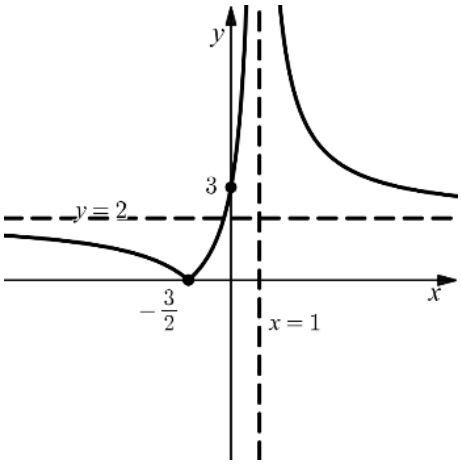


7) a)

Consider $y = \frac{2x+3}{x-1}$:

When $x = 0$, $y = -3$, when $y = 0$, $x = -\frac{3}{2}$

Asymptotes: $x = 1$, $y = 2$

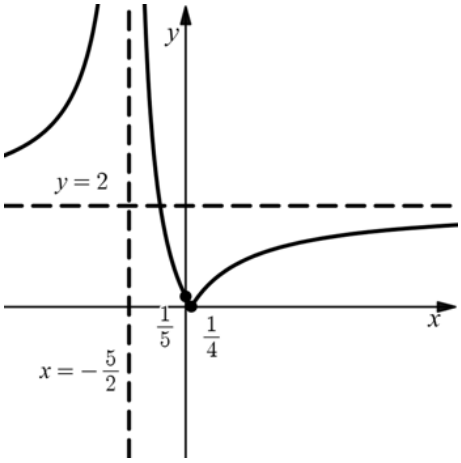


7) b)

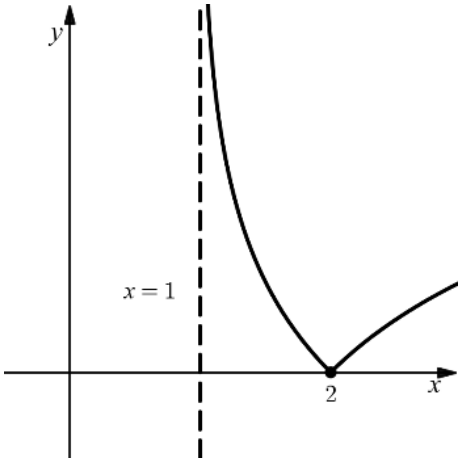
Consider $y = \frac{1-4x}{2x+5}$:

When $x = 0$, $y = \frac{1}{5}$, when $y = 0$, $x = \frac{1}{4}$

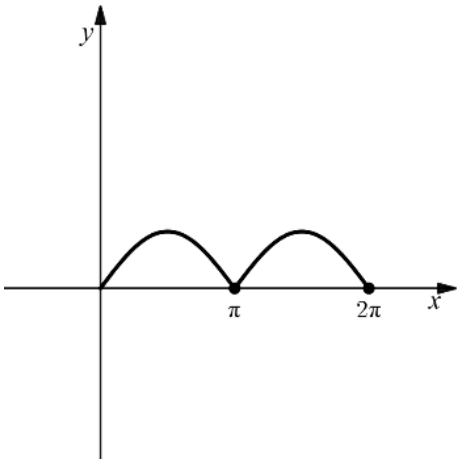
Asymptotes: $x = -\frac{5}{2}$, $y = |-2| = 2$



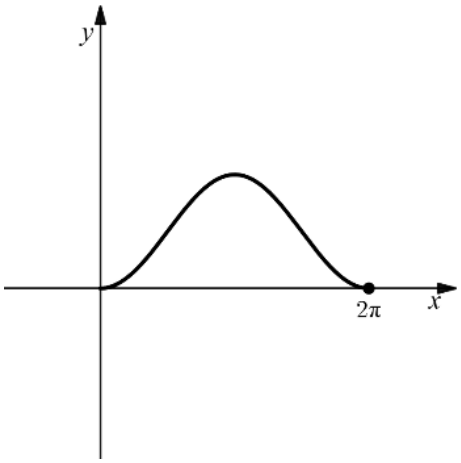
8) a)



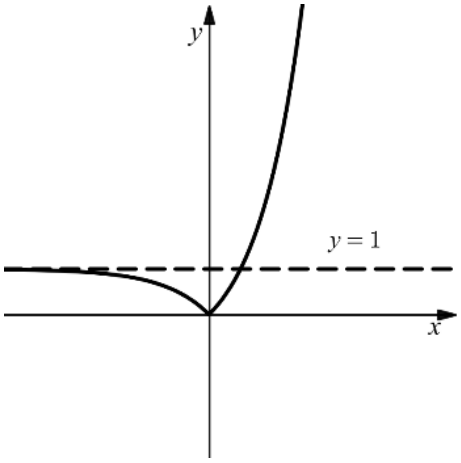
8) b)



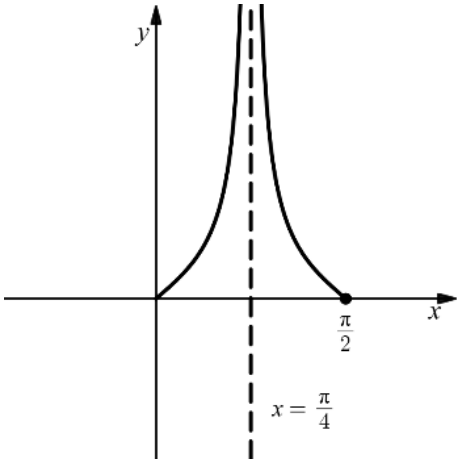
8) c)



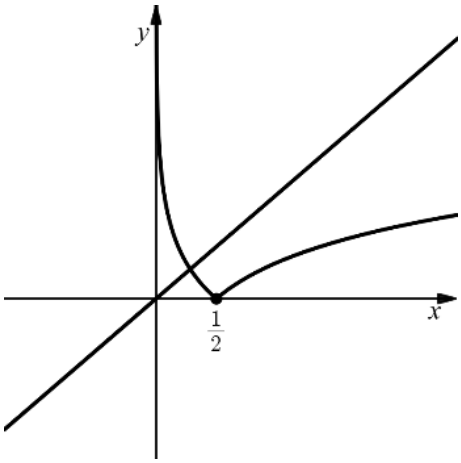
8) d)



8) e)

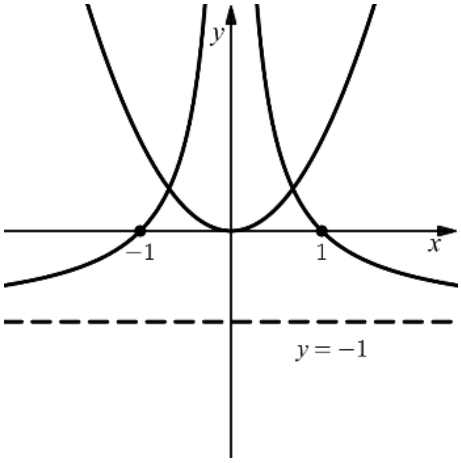


9) a)



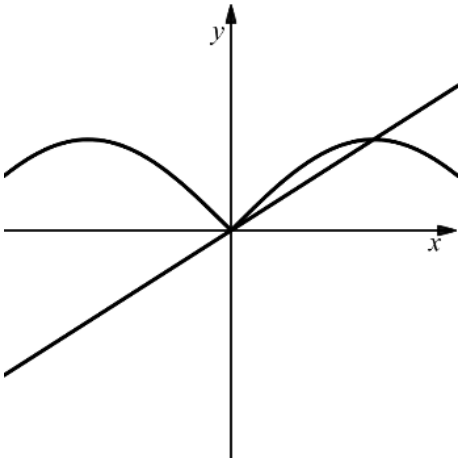
∴ 1 solution.

9) b)



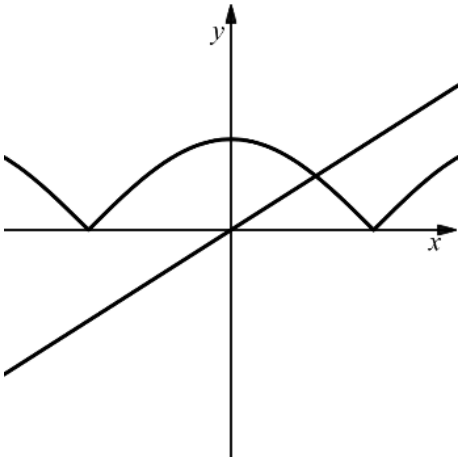
∴ 2 solutions.

9) c)



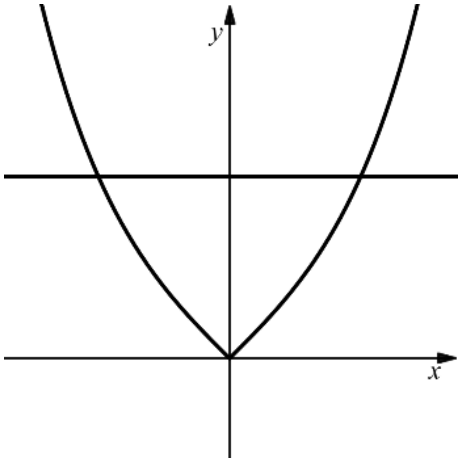
∴ 2 solutions.

9) d)



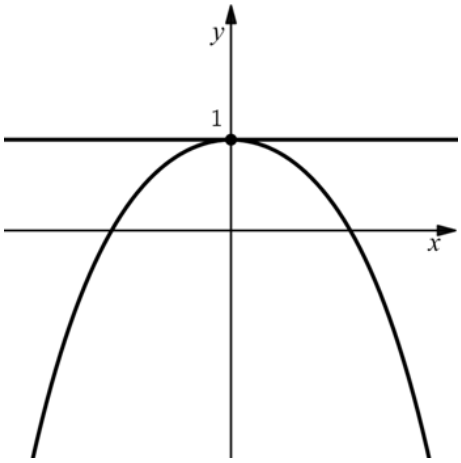
∴ 1 solution.

10) a)



∴ 2 solutions.

10) b)



∴ 1 solution.

11) a)

$\sinh x = 1 \Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

Then by symmetry, $-\ln\left(1 + \sqrt{2}\right)$

$\therefore x = \ln\left(1 + \sqrt{2}\right)$ or $x = -\ln\left(1 + \sqrt{2}\right)$

11) b)

$\cosh x - 2 = 1 \Rightarrow \cosh x = 3 \Rightarrow x = \ln\left(3 + \sqrt{3^2 - 1}\right) = \ln\left(3 + 2\sqrt{2}\right)$

Then by symmetry, $\ln\left(3 - 2\sqrt{2}\right)$

$\cosh x - 2 = -1 \Rightarrow \cosh x = 1 \Rightarrow x = \ln\left(1 + \sqrt{1^2 - 1}\right) = 0$

$\therefore x = \ln\left(3 + 2\sqrt{2}\right)$, $x = \ln\left(3 - 2\sqrt{2}\right)$ or $x = 0$

12)

From the asymptote: $x = -\frac{1}{3}$

$c\left(-\frac{1}{3}\right) + d = 0 \Rightarrow d = \frac{c}{3}$

From asymptote: $y = \frac{2}{3}$

$\left|\frac{a}{c}\right| = \frac{2}{3}$

So take smallest integers: $c = 3$, $d = 1$, $a = 2$

So far: $y = \left|\frac{2x+b}{3x+1}\right|$

x-intercept is 2 so:

$2 \times 2 + b = 0 \Rightarrow b = -4$

$\therefore y = \left|\frac{2x-4}{3x+1}\right|$

Check y-intercept matches:

When $x = 0$, $y = \left|\frac{2\times0-4}{3\times0+1}\right| = 4$ as required

$\therefore a = 2$, $b = -4$ $c = 3$ and $d = 1$